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ABSTRACT

AI – Powered Personalized Learning System

The proposed system is an AI-powered personalized learning platform that integrates multimodal document processing, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and student performance analysis. Inspired by the selected multimodal RAG reference work, the system processes educational materials in formats such as PDF, DOCX, PPTX, and images using document extraction and Optical Character Recognition (OCR). The extracted content is divided into smaller chunks, converted into semantic embeddings, and stored in a vector database for efficient retrieval. When students ask questions, relevant information from their uploaded materials is retrieved and provided as context to a locally deployed LLM for generating grounded responses. The system further extends this approach by incorporating test and grade analysis to identify topic-wise strengths and weaknesses and support personalized assessment. By connecting study materials with student performance, the proposed system aims to provide a context-aware, multimodal, and personalized learning environment. The goal of the project is to reduce dependence on Generic AI and instead switch to a more personalized and customized AI that helps with targeted learning. **Keywords – Retrieval-Augmented Generation, Large Language Models, Optical Recognition, Semantic Embeddings.**

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